

Year 7 • Chapter 8 • Area and volume: answers (teacher copy)

8.01 Units of measurement (booklet p. 3)

WE DO examples (teacher)

- Column 1, E1: $5.7 \text{ t} = \underline{\quad} \text{ kg} \rightarrow 5\,700 \text{ kg}$
- Column 1, E2: $27\,000 \text{ kg} = \underline{\quad} \text{ t} \rightarrow 27 \text{ t}$
- Column 2, E1: $32 \text{ kL} = \underline{\quad} \text{ L} \rightarrow \text{kL to L: } \times 1\,000; 32 \times 1\,000 = 32\,000 \text{ L}$
- Column 2, E2: $2 \text{ kg} = \underline{\quad} \text{ g} \rightarrow \text{kg to g: } \times 1\,000; 2 \times 1\,000 = 2\,000 \text{ g}$
- Column 3, E1: Add 7 m and 910 cm. Give the answer in metres. $\rightarrow 910 \text{ cm} = 9.1 \text{ m}; 7 + 9.1 = 16.1 \text{ m}$
- Column 3, E2: A 8 L jug has 461 mL poured out. How much is left, in mL? $\rightarrow 8 \text{ L} = 8\,000 \text{ mL}; 8\,000 - 461 = 7\,539 \text{ mL}$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|--------------|---------------|--------------|
| 1. 57 000 mm | 17. 1 500 mm | 33. 6 000 g |
| 2. 5.5 m | 18. 490 cm | 34. 38 kL |
| 3. 4 800 m | 19. 36 kL | 35. 20 L |
| 4. 270 mm | 20. 7 kg | 36. 2 600 mL |
| 5. 560 mm | 21. 57 000 mL | 37. 4.5 m |
| 6. 46 000 L | 22. 56 t | 38. 46 cm |
| 7. 19 000 mm | 23. 1.4 L | 39. 4 000 kg |
| 8. 24 L | 24. 44 mm | 40. 10.18 m |
| 9. 48 m | 25. 100 cm | 41. 4465 mL |
| 10. 4.8 cm | 26. 44 m | 42. 5.51 m |
| 11. 2.2 cm | 27. 210 mm | 43. 7504 mL |
| 12. 3 200 mm | 28. 35 000 m | 44. 12.25 m |
| 13. 8 kL | 29. 3.8 kg | 45. 3095 mL |
| 14. 42 t | 30. 10 km | 46. 6.88 m |
| 15. 44 kL | 31. 23 L | 47. 6601 mL |
| 16. 9 t | 32. 4 km | |

8.02 Perimeter (booklet p. 4)

WE DO examples (teacher)

- Column 1, E1: 21 m by 9 m rectangle $\rightarrow 60 \text{ m}$
- Column 1, E2: 21 m by 8 m rectangle $\rightarrow 58 \text{ m}$
- Column 2, E1: (diagram) $\rightarrow 22 + 17 + 22 + 17; = 78 \text{ cm}$
- Column 2, E2: (diagram) $\rightarrow 27 + 15 + 27 + 15; = 84 \text{ cm}$
- Column 3, E1: A rectangle has perimeter 66 m and length 25 m. Find its width. $\rightarrow 2 \times 25 = 50; 66 - 50 = 16; \text{width} = 16 \div 2 = 8 \text{ m}$
- Column 3, E2: A square has perimeter 32 cm. Find the side length. $\rightarrow 4 \text{ equal sides}; 32 \div 4 = 8 \text{ cm}$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|----------|----------|-----------|
| 1. 20 m | 11. 24 m | 21. 48 cm |
| 2. 54 m | 12. 54 m | 22. 70 cm |
| 3. 26 m | 13. 44 m | 23. 82 cm |
| 4. 38 m | 14. 42 m | 24. 18 m |
| 5. 72 m | 15. 66 m | 25. 10 cm |
| 6. 36 m | 16. 42 m | 26. 6 m |
| 7. 56 m | 17. 52 m | 27. 20 cm |
| 8. 52 m | 18. 36 m | 28. 5 m |
| 9. 56 m | 19. 80 m | 29. 19 cm |
| 10. 62 m | 20. 76 m | |

8.03 Circumference of a circle (booklet p. 5)

WE DO examples (teacher)

- Column 1, E1: $d = 3 \text{ cm} \rightarrow 9.4 \text{ cm}$
- Column 1, E2: $d = 38 \text{ cm} \rightarrow 119.4 \text{ cm}$
- Column 2, E1: (diagram) $\rightarrow C = 2\pi r = 2 \times \pi \times 6; = 37.7 \text{ m}$
- Column 2, E2: (diagram) $\rightarrow C = \pi d = \pi \times 10; = 31.4 \text{ m}$

- Column 3, E1: A circle has circumference 53 cm. Find its diameter (1 d.p.). $\rightarrow C = \pi d$, so $d = C \div \pi; d = 53 \div \pi; = 16.9 \text{ cm}$
- Column 3, E2: Find the perimeter of a semicircle with diameter 19 cm (1 d.p.). $\rightarrow \text{curve: } \pi \times 19 \div 2 = 29.8$; add the diameter: $29.8 + 19; = 48.8 \text{ cm}$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|--------------|--------------|--------------|
| 1. 66.0 cm | 14. 62.8 cm | 27. 69.1 cm |
| 2. 84.8 cm | 15. 91.1 cm | 28. 31.4 cm |
| 3. 72.3 cm | 16. 125.7 cm | 29. 113.1 cm |
| 4. 106.8 cm | 17. 100.5 cm | 30. 31.4 m |
| 5. 34.6 cm | 18. 116.2 cm | 31. 44.0 m |
| 6. 47.1 cm | 19. 25.1 cm | 32. 44.0 m |
| 7. 59.7 cm | 20. 37.7 cm | 33. 22.9 cm |
| 8. 56.5 cm | 21. 50.3 cm | 34. 43.7 cm |
| 9. 40.8 cm | 22. 75.4 cm | 35. 18.1 cm |
| 10. 28.3 cm | 23. 103.7 cm | 36. 15.4 cm |
| 11. 18.8 cm | 24. 22.0 cm | 37. 22.0 cm |
| 12. 15.7 cm | 25. 12.6 cm | |
| 13. 122.5 cm | 26. 6.3 cm | |

8.04 Area units (booklet p. 6)

WE DO examples (teacher)

- Column 1, E1: $19 \text{ ha} = \underline{\quad} \text{ m}^2 \rightarrow 190\,000 \text{ m}^2$
- Column 1, E2: $16 \text{ cm}^2 = \underline{\quad} \text{ mm}^2 \rightarrow 1\,600 \text{ mm}^2$
- Column 2, E1: $2.5 \text{ km}^2 = \underline{\quad} \text{ m}^2 \rightarrow 1 \text{ km}^2 = 1\,000 \times 1\,000 = 1\,000\,000 \text{ m}^2; 2.5 \times 1\,000\,000 = 2\,500\,000 \text{ m}^2$
- Column 2, E2: $2 \text{ m}^2 = \underline{\quad} \text{ cm}^2 \rightarrow 1 \text{ m}^2 = 100 \times 100 = 10\,000 \text{ cm}^2; 2 \times 10\,000 = 20\,000 \text{ cm}^2$
- Column 3, E1: A paddock is 200 m by 100 m. Find its area in hectares. $\rightarrow A = 200 \times 100 = 20\,000 \text{ m}^2; \div 10\,000: 2 \text{ ha}$
- Column 3, E2: A paddock is 800 m by 100 m. Find its area in hectares. $\rightarrow A = 800 \times 100 = 80\,000 \text{ m}^2; \div 10\,000: 8 \text{ ha}$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|-----------------------------|-------------------------------|-------------------------------|
| 1. 7 m^2 | 16. 12 ha | 31. $25\,000 \text{ cm}^2$ |
| 2. $170\,000 \text{ cm}^2$ | 17. 30 cm^2 | 32. $8\,000\,000 \text{ m}^2$ |
| 3. 25 cm^2 | 18. 3 m^2 | 33. $4\,000\,000 \text{ m}^2$ |
| 4. 7 ha | 19. $50\,000 \text{ m}^2$ | 34. $60\,000 \text{ cm}^2$ |
| 5. 500 mm^2 | 20. 18 m^2 | 35. $3\,000\,000 \text{ m}^2$ |
| 6. 19 m^2 | 21. $210\,000 \text{ m}^2$ | 36. 450 mm^2 |
| 7. 19 cm^2 | 22. $70\,000 \text{ cm}^2$ | 37. $90\,000 \text{ cm}^2$ |
| 8. 12 cm^2 | 23. 22 m^2 | 38. 16 ha |
| 9. 8 ha | 24. $160\,000 \text{ m}^2$ | 39. 40 ha |
| 10. $140\,000 \text{ m}^2$ | 25. 14 ha | 40. 12 ha |
| 11. $270\,000 \text{ cm}^2$ | 26. $20\,000 \text{ m}^2$ | 41. 6 ha |
| 12. $170\,000 \text{ m}^2$ | 27. 27 cm^2 | 42. 45 ha |
| 13. $80\,000 \text{ m}^2$ | 28. 2 m^2 | 43. 10 ha |
| 14. 23 cm^2 | 29. 700 mm^2 | 44. 35 ha |
| 15. $180\,000 \text{ cm}^2$ | 30. $5\,500\,000 \text{ m}^2$ | |

8.05 Area of squares and rectangles

(booklet p. 7)

WE DO examples (teacher)

- Column 1, E1: 3 m by 3 m $\rightarrow 9 \text{ m}^2$
- Column 1, E2: 7 m by 7 m $\rightarrow 49 \text{ m}^2$
- Column 2, E1: (diagram) $\rightarrow A = lw = 15 \times 13; = 195 \text{ cm}^2$
- Column 2, E2: (diagram) $\rightarrow A = lw = 8 \times 3; = 24 \text{ cm}^2$

5. Column 3, E1: A rectangle has area 8 m^2 and width 2 m. Find its length. $\rightarrow A = lw$, so $8 = l \times 2$; $l = 8 \div 2 = 4 \text{ m}$
6. Column 3, E2: A square has sides of 7.5 cm. Find its area. $\rightarrow A = s^2 = 7.5 \times 7.5 = 56.25 \text{ cm}^2$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|-----------------------|-----------------------|---------------------------|
| 1. 75 m^2 | 15. 130 m^2 | 29. 27 m^2 |
| 2. 18 m^2 | 16. 8 m^2 | 30. 39 m^2 |
| 3. 30 m^2 | 17. 70 m^2 | 31. 128 cm^2 |
| 4. 150 m^2 | 18. 80 m^2 | 32. 27 cm^2 |
| 5. 36 m^2 | 19. 180 m^2 | 33. 88 cm^2 |
| 6. 15 m^2 | 20. 182 m^2 | 34. 17 m |
| 7. 195 m^2 | 21. 42 m^2 | 35. 210.25 cm^2 |
| 8. 100 m^2 | 22. 48 m^2 | 36. 13 m |
| 9. 45 m^2 | 23. 14 m^2 | 37. 90.25 cm^2 |
| 10. 16 m^2 | 24. 117 m^2 | 38. 5 m |
| 11. 48 m^2 | 25. 154 m^2 | 39. 30.25 cm^2 |
| 12. 225 m^2 | 26. 63 m^2 | 40. 7 m |
| 13. 140 m^2 | 27. 25 m^2 | 41. 12.25 cm^2 |
| 14. 6 m^2 | 28. 54 m^2 | |

8.06 Area of a triangle (booklet p. 8)

WE DO examples (teacher)

- Column 1, E1: $b = 4 \text{ cm}$, $h = 20 \text{ cm} \rightarrow 40 \text{ cm}^2$
- Column 1, E2: $b = 14 \text{ cm}$, $h = 2 \text{ cm} \rightarrow 14 \text{ cm}^2$
- Column 2, E1: (diagram) $\rightarrow A = \frac{1}{2} \times 8 \times 7 = 28 \text{ cm}^2$
- Column 2, E2: (diagram) $\rightarrow A = \frac{1}{2} \times 13 \times 3 = 19.5 \text{ cm}^2$
- Column 3, E1: A triangle has area 42 cm^2 and base 6 cm. Find its height. $\rightarrow 42 = \frac{1}{2} \times 6 \times h$; $42 = 3 \times h$; $h = 42 \div 3 = 14 \text{ cm}$
- Column 3, E2: A triangle has area 104 cm^2 and base 16 cm. Find its height. $\rightarrow 104 = \frac{1}{2} \times 16 \times h$; $104 = 8 \times h$; $h = 104 \div 8 = 13 \text{ cm}$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|------------------------|------------------------|------------------------|
| 1. 56 cm^2 | 14. 8 cm^2 | 27. 90 cm^2 |
| 2. 15 cm^2 | 15. 9 cm^2 | 28. 104 cm^2 |
| 3. 18 cm^2 | 16. 102 cm^2 | 29. 24 cm^2 |
| 4. 126 cm^2 | 17. 78 cm^2 | 30. 56 cm^2 |
| 5. 90 cm^2 | 18. 55 cm^2 | 31. 48 cm^2 |
| 6. 88 cm^2 | 19. 32 cm^2 | 32. 36 cm^2 |
| 7. 56 cm^2 | 20. 54 cm^2 | 33. 55 cm^2 |
| 8. 56 cm^2 | 21. 22 cm^2 | 34. 15 cm |
| 9. 8 cm^2 | 22. 27 cm^2 | 35. 8 cm |
| 10. 45 cm^2 | 23. 81 cm^2 | 36. 6 cm |
| 11. 21 cm^2 | 24. 48 cm^2 | 37. 16 cm |
| 12. 120 cm^2 | 25. 30 cm^2 | 38. 6 cm |
| 13. 11 cm^2 | 26. 36 cm^2 | 39. 4 cm |

8.07 Area of a parallelogram (booklet p. 9)

WE DO examples (teacher)

- Column 1, E1: $b = 8 \text{ m}$, $h = 5 \text{ m} \rightarrow 40 \text{ m}^2$
- Column 1, E2: $b = 9 \text{ m}$, $h = 7 \text{ m} \rightarrow 63 \text{ m}^2$
- Column 2, E1: (diagram) $\rightarrow A = bh = 19 \times 7 = 133 \text{ cm}^2$
- Column 2, E2: (diagram) $\rightarrow A = bh = 7 \times 12 = 84 \text{ cm}^2$
- Column 3, E1: A parallelogram has area 104 m^2 and height 8 m. Find its base. $\rightarrow 104 = b \times 8$; $b = 104 \div 8 = 13 \text{ m}$
- Column 3, E2: A parallelogram has area 98 m^2 and height 7 m. Find its base. $\rightarrow 98 = b \times 7$; $b = 98 \div 7 = 14 \text{ m}$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|-----------------------|-----------------------|-----------------------|
| 1. 36 m^2 | 13. 55 m^2 | 25. 40 m^2 |
| 2. 285 m^2 | 14. 36 m^2 | 26. 36 m^2 |
| 3. 21 m^2 | 15. 192 m^2 | 27. 52 m^2 |
| 4. 135 m^2 | 16. 45 m^2 | 28. 90 m^2 |
| 5. 210 m^2 | 17. 49 m^2 | 29. 100 m^2 |
| 6. 38 m^2 | 18. 90 m^2 | 30. 60 cm^2 |
| 7. 90 m^2 | 19. 54 m^2 | 31. 80 cm^2 |
| 8. 180 m^2 | 20. 30 m^2 | 32. 70 cm^2 |
| 9. 144 m^2 | 21. 60 m^2 | 33. 7 m |
| 10. 9 m^2 | 22. 96 m^2 | 34. 15 m |
| 11. 209 m^2 | 23. 152 m^2 | 35. 6 m |
| 12. 55 m^2 | 24. 154 m^2 | 36. 6 m |
| | | 37. 16 m |

8.08 Area of composite shapes (booklet p. 10)

WE DO examples (teacher)

- Column 1, E1: 9×4 and $6 \times 7 \text{ (m)} \rightarrow 78 \text{ m}^2$
- Column 1, E2: 5×6 and $10 \times 5 \text{ (m)} \rightarrow 80 \text{ m}^2$
- Column 2, E1: (diagram) $\rightarrow 4 \times 8 = 32$; $6 \times 4 = 24$; total: 56 cm^2
- Column 2, E2: (diagram) $\rightarrow 7 \times 8 = 56$; $3 \times 4 = 12$; total: 68 cm^2
- Column 3, E1: A $15 \text{ cm} \times 12 \text{ cm}$ rectangle has a $3 \text{ cm} \times 3 \text{ cm}$ hole cut out. Find the area left. $\rightarrow \text{big: } 15 \times 12 = 180$; hole: $3 \times 3 = 9$; $180 - 9 = 171 \text{ cm}^2$
- Column 3, E2: A house wall is a $18 \text{ m} \times 9 \text{ m}$ rectangle with a triangle on top (base 18 m, height 3 m). Find its area. $\rightarrow \text{rectangle: } 18 \times 9 = 162$; triangle: $\frac{1}{2} \times 18 \times 3 = 27$; total: 189 m^2

Questions (in order: Easy, Medium, Challenging)

- | | | |
|-----------------------|-----------------------|------------------------|
| 1. 75 m^2 | 13. 80 m^2 | 25. 42 m^2 |
| 2. 52 m^2 | 14. 76 m^2 | 26. 46 m^2 |
| 3. 106 m^2 | 15. 80 m^2 | 27. 30 m^2 |
| 4. 93 m^2 | 16. 25 m^2 | 28. 49 m^2 |
| 5. 49 m^2 | 17. 50 m^2 | 29. 78 m^2 |
| 6. 53 m^2 | 18. 107 m^2 | 30. 60 cm^2 |
| 7. 156 m^2 | 19. 92 m^2 | 31. 50 cm^2 |
| 8. 24 m^2 | 20. 118 m^2 | 32. 103 cm^2 |
| 9. 51 m^2 | 21. 130 m^2 | 33. 169 m^2 |
| 10. 103 m^2 | 22. 42 m^2 | 34. 174 cm^2 |
| 11. 32 m^2 | 23. 78 m^2 | 35. 248 m^2 |
| 12. 78 m^2 | 24. 37 m^2 | |

8.09 Prisms, cross-sections and views

(booklet p. 11)

WE DO examples (teacher)

- Column 1, E1: triangular prism $\rightarrow$ triangle
- Column 1, E2: rectangular prism $\rightarrow$ rectangle
- Column 2, E1: cross-section: triangle $\rightarrow n = 3$: $F = 3 + 2 = 5$; $V = 2 \times 3 = 6$, $E = 3 \times 3 = 9$
- Column 2, E2: cross-section: square $\rightarrow n = 4$: $F = 4 + 2 = 6$; $V = 2 \times 4 = 8$, $E = 3 \times 4 = 12$
- Column 3, E1: a cylinder standing up $\rightarrow$ front view: rectangle; top view: circle
- Column 3, E2: a square pyramid $\rightarrow$ front view: triangle; top view: square with diagonals

Questions (in order: Easy, Medium, Challenging)

- | | | |
|-------------|------------|------------|
| 1. pentagon | 2. hexagon | 3. octagon |
|-------------|------------|------------|

- | | | |
|---------------|--------------------------|----------------------|
| 4. square | 17. heptagon | 28. front: |
| 5. circle | 18. dodecagon (12 sides) | rectangle; top: |
| 6. triangle | 19. circle (a ring) | rectangle with |
| 7. rectangle | 20. rectangle | a line down the |
| 8. hexagon | 21. square | middle |
| 9. square | 22. F 7, V 10, E 15 | 29. front: triangle; |
| 10. triangle | 23. F 8, V 12, E 18 | top: circle with |
| 11. circle | 24. F 9, V 14, E 21 | a dot in the |
| 12. rectangle | 25. F 10, V 16, E 24 | centre |
| 13. hexagon | 26. F 12, V 20, E 30 | 30. front: square; |
| 14. octagon | 27. F 14, V 24, E 36 | top: square |
| 15. decagon | | 31. front: tall |
| 16. rectangle | | rectangle; top: |
| | | small rectangle |

8.10 Volume units (booklet p. 12)

WE DO examples (teacher)

- Column 1, E1: $140 \text{ cm}^3 = \underline{\quad} \text{ mL} \rightarrow 140 \text{ mL}$
- Column 1, E2: $36\,000 \text{ cm}^3 = \underline{\quad} \text{ L} \rightarrow 36 \text{ L}$
- Column 2, E1: $4 \text{ m}^3 = \underline{\quad} \text{ cm}^3 \rightarrow 1 \text{ m}^3 = 100 \times 100 \times 100 = 1\,000\,000 \text{ cm}^3$; $4 \times 1\,000\,000 = 4\,000\,000 \text{ cm}^3$
- Column 2, E2: $9 \text{ cm}^3 = \underline{\quad} \text{ mm}^3 \rightarrow 1 \text{ cm}^3 = 10 \times 10 \times 10 = 1\,000 \text{ mm}^3$; $9 \times 1\,000 = 9\,000 \text{ mm}^3$
- Column 3, E1: A tank holds 8.5 m^3 of water. How many litres is that? $\rightarrow 1 \text{ m}^3 = 1\,000 \text{ L}$; $8.5 \times 1\,000 = 8\,500 \text{ L}$
- Column 3, E2: A tank holds 1 m^3 of water. How many litres is that? $\rightarrow 1 \text{ m}^3 = 1\,000 \text{ L}$; $1 \times 1\,000 = 1\,000 \text{ L}$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|----------------------------|--------------------------------|--------------------------------|
| 1. 560 mL | 17. $49\,000 \text{ mm}^3$ | 33. $6\,000\,000 \text{ cm}^3$ |
| 2. 22 L | 18. 30 L | 34. $6\,000 \text{ mm}^3$ |
| 3. 190 mL | 19. $12\,000 \text{ mm}^3$ | 35. $9\,000\,000 \text{ cm}^3$ |
| 4. 340 mL | 20. 38 L | 36. $5\,000 \text{ mm}^3$ |
| 5. 44 L | 21. 20 L | 37. $4\,000 \text{ mm}^3$ |
| 6. 42 L | 22. 50 L | 38. $8\,000 \text{ mm}^3$ |
| 7. 4 L | 23. 540 mL | 39. $7\,000\,000 \text{ cm}^3$ |
| 8. 250 mL | 24. 54 L | 40. 8 000 L |
| 9. $31\,000 \text{ mm}^3$ | 25. 510 mL | 41. 3 000 L |
| 10. 59 L | 26. 370 mL | 42. 4 000 L |
| 11. 51 L | 27. 33 L | 43. 6 500 L |
| 12. 290 mL | 28. 52 L | 44. 9 500 L |
| 13. 8 L | 29. 550 mL | 45. 2 500 L |
| 14. 260 mL | 30. $3\,000 \text{ mm}^3$ | 46. 7 500 L |
| 15. $8\,000 \text{ mm}^3$ | 31. $3\,000\,000 \text{ cm}^3$ | 47. 1 500 L |
| 16. $22\,000 \text{ mm}^3$ | 32. $7\,000 \text{ mm}^3$ | |

8.11 Volume of a rectangular prism (booklet p. 13)

WE DO examples (teacher)

- Column 1, E1: $4 \text{ cm} \times 9 \text{ cm} \times 2 \text{ cm} \rightarrow 72 \text{ cm}^3$
- Column 1, E2: $8 \text{ cm} \times 9 \text{ cm} \times 7 \text{ cm} \rightarrow 504 \text{ cm}^3$
- Column 2, E1: (diagram) $\rightarrow V = lwh = 5 \times 6 \times 5 = 150 \text{ cm}^3$
- Column 2, E2: (diagram) $\rightarrow V = lwh = 9 \times 9 \times 5 = 405 \text{ cm}^3$
- Column 3, E1: A box has volume 192 cm^3 , length 12 cm and width 8 cm. Find its height. $\rightarrow 192 = 12 \times 8 \times h$; $192 = 96 \times h$; $h = 192 \div 96 = 2 \text{ cm}$
- Column 3, E2: A box has volume 84 cm^3 , length 7 cm and width 2 cm. Find its height. $\rightarrow 84 = 7 \times 2 \times h$; $84 = 14 \times h$; $h = 84 \div 14 = 6 \text{ cm}$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|----------------------|----------------------|-----------------------|
| 1. 72 cm^3 | 2. 30 cm^3 | 3. 120 cm^3 |
|----------------------|----------------------|-----------------------|

- | | | |
|------------------------|------------------------|------------------------|
| 4. 72 cm^3 | 16. 693 cm^3 | 28. 72 cm^3 |
| 5. 120 cm^3 | 17. 693 cm^3 | 29. 70 cm^3 |
| 6. 576 cm^3 | 18. 192 cm^3 | 30. 432 cm^3 |
| 7. 729 cm^3 | 19. 192 cm^3 | 31. 66 cm^3 |
| 8. 144 cm^3 | 20. 392 cm^3 | 32. 360 cm^3 |
| 9. 288 cm^3 | 21. 80 cm^3 | 33. 260 cm^3 |
| 10. 196 cm^3 | 22. 300 cm^3 | 34. 7 cm |
| 11. 60 cm^3 | 23. 540 cm^3 | 35. 6 cm |
| 12. 144 cm^3 | 24. 243 cm^3 | 36. 5 cm |
| 13. 448 cm^3 | 25. 28 cm^3 | 37. 9 cm |
| 14. 180 cm^3 | 26. 128 cm^3 | 38. 6 cm |
| 15. 140 cm^3 | 27. 240 cm^3 | |

8.12 Volume of a prism (booklet p. 14)

WE DO examples (teacher)

- Column 1, E1: $A = 19 \text{ cm}^2$, $l = 4 \text{ cm} \rightarrow 76 \text{ cm}^3$
- Column 1, E2: $A = 5 \text{ cm}^2$, $l = 8 \text{ cm} \rightarrow 40 \text{ cm}^3$
- Column 2, E1: (diagram) $\rightarrow A = \frac{1}{2} \times 9 \times 8 = 36$; $V = 36 \times 20 = 720 \text{ cm}^3$
- Column 2, E2: (diagram) $\rightarrow A = \frac{1}{2} \times 8 \times 5 = 20$; $V = 20 \times 13 = 260 \text{ cm}^3$
- Column 3, E1: A prism has volume 330 m^3 and cross-section area 22 m^2 . Find its length. $\rightarrow 330 = 22 \times \text{length}$; $\text{length} = 330 \div 22 = 15 \text{ m}$
- Column 3, E2: A prism has volume 95 m^3 and cross-section area 19 m^2 . Find its length. $\rightarrow 95 = 19 \times \text{length}$; $\text{length} = 95 \div 19 = 5 \text{ m}$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|------------------------|------------------------|------------------------|
| 1. 27 cm^3 | 13. 156 cm^3 | 25. 495 cm^3 |
| 2. 60 cm^3 | 14. 198 cm^3 | 26. 32 cm^3 |
| 3. 264 cm^3 | 15. 205 cm^3 | 27. 357 cm^3 |
| 4. 170 cm^3 | 16. 125 cm^3 | 28. 140 cm^3 |
| 5. 84 cm^3 | 17. 102 cm^3 | 29. 300 cm^3 |
| 6. 144 cm^3 | 18. 266 cm^3 | 30. 228 cm^3 |
| 7. 330 cm^3 | 19. 64 cm^3 | 31. 520 cm^3 |
| 8. 90 cm^3 | 20. 234 cm^3 | 32. 9 m |
| 9. 312 cm^3 | 21. 560 cm^3 | 33. 9 m |
| 10. 190 cm^3 | 22. 147 cm^3 | 34. 3 m |
| 11. 12 cm^3 | 23. 24 cm^3 | 35. 5 m |
| 12. 203 cm^3 | 24. 528 cm^3 | 36. 11 m |

8.13 Capacity (booklet p. 15)

WE DO examples (teacher)

- Column 1, E1: $1\,500 \text{ mL} = \underline{\quad} \text{ L} \rightarrow 1.5 \text{ L}$
- Column 1, E2: $5 \text{ kL} = \underline{\quad} \text{ L} \rightarrow 5\,000 \text{ L}$
- Column 2, E1: a box $30 \text{ cm} \times 10 \text{ cm} \times 15 \text{ cm} \rightarrow V = 30 \times 10 \times 15 = 4\,500 \text{ cm}^3$; $\div 1\,000 = 4.5 \text{ L}$
- Column 2, E2: a box $20 \text{ cm} \times 10 \text{ cm} \times 15 \text{ cm} \rightarrow V = 20 \times 10 \times 15 = 3\,000 \text{ cm}^3$; $\div 1\,000 = 3 \text{ L}$
- Column 3, E1: How many 500 mL cups can be filled from 3 L of juice? $\rightarrow 3 \text{ L} = 3\,000 \text{ mL}$; $3\,000 \div 500 = 6$; 6 cups
- Column 3, E2: How many 750 mL cups can be filled from 4.5 L of juice? $\rightarrow 4.5 \text{ L} = 4\,500 \text{ mL}$; $4\,500 \div 750 = 6$; 6 cups

Questions (in order: Easy, Medium, Challenging)

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|--------------|-------------|--------------|
| 1. 11 000 mL | 5. 8 500 mL | 9. 9 L |
| 2. 0.25 kL | 6. 18 kL | 10. 14 kL |
| 3. 2 500 L | 7. 16.5 L | 11. 11 500 L |
| 4. 1 kL | 8. 1 000 mL | 12. 19 kL |

13. 500 L	19. 2 000 L	25. 0.75 kL	31. 3 L	37. 40 L	43. 30 cups
14. 5.5 L	20. 1.75 kL	26. 31 kL	32. 4 L	38. 60 L	44. 8 cups
15. 1 500 mL	21. 1 750 L	27. 28 L	33. 10 L	39. 37.5 L	45. 2 cups
16. 35 L	22. 7 L	28. 6 L	34. 2 L	40. 50 L	46. 5 cups
17. 21 000 L	23. 31 000 L	29. 8.5 kL	35. 15 L	41. 36 cups	
18. 8 kL	24. 29 kL	30. 4 500 L	36. 20 L	42. 3 cups	